Appendix 1

Museums: A Conversational Encyclopedia of Research

1. What Museums Are and Why They Matter

Museums are institutions that collect, preserve, research, and exhibit objects of historical, cultural, scientific, or artistic value. According to Wikipedia, they exist to safeguard and interpret the tangible and intangible heritage of humanity. The International Council of Museums (ICOM) defines a museum as *"a not-for-profit, permanent institution in the service of society that researches, collects, conserves, interprets and exhibits tangible and intangible heritage. Open to the public, accessible and inclusive, museums foster diversity and sustainability. They operate and communicate ethically, professionally and with the participation of communities, offering varied experiences for education, enjoyment, reflection and knowledge."*

Museums are often described as public-facing institutions, places of education, engagement, and identity. They connect people with culture, ideas, and each other. They are not only buildings full of artifacts but also living spaces for learning, reflection, and community. They help societies tell their stories, celebrate creativity, and preserve heritage for the future. Museums matter because they influence people in many ways, cognitively, emotionally, socially, and even economically. Visitors can experience the stimulation of seeing real works of art, the calm of a quiet gallery, or the excitement of an interactive exhibit. Studies show that museum visits can reduce stress and improve well-being. Research published in Museums & Social Issues reported measurable reductions in psychological and physiological stress indicators after art-museum visits.

The Barber Institute of Fine Arts has shown that museums contribute to well-being through social interaction, aesthetic experiences, and the emotional journeys they create. In short, museums serve as anchors of education, culture, and emotional connection, providing experiences that foster knowledge, curiosity, and belonging.

2. How Museums Began: A Short History

Early Beginnings: Collections Before Museums

The story of museums begins long before the word museum existed. Collecting has been part of human behavior since ancient times. Across Africa, Arabia, and Asia, people assembled objects for prestige, spirituality, and learning, long before formal museums appeared in Europe.

One of the oldest known collections dates to around 530 BCE in the ancient Sumerian city of Uruk (modern-day Iraq). This early collection, curated by Princess Ennigaldi, daughter of Nabonidus, the last king of the Neo-Babylonian Empire, is often considered the first museum in history. It displayed ancient artifacts with labels describing their origins, showing a remarkable early awareness of historical preservation.

The Term and Concept of "Museum"

The word museum originates from the Greek "mouseion", meaning "seat of the Muses." In ancient Greece, this referred to a place dedicated to the Muses, goddesses of the arts and sciences. These early "museums" were spaces for learning and contemplation.

One of the most famous examples was the Mouseion of Alexandria, founded in the 3rd century BCE by Ptolemy Soter (305–283 BCE). It combined learning, research, and collecting in one institution. The Mouseion housed scholars such as Euclid and Heraclitus, and contained the largest library in the ancient world, along with lecture halls, gardens, and living quarters. It functioned as both a university and a research center. Sadly, the temple and its vast collections were destroyed, probably by fire, during later conflicts in Alexandria.

The Evolution of Collecting Traditions

Between 900 and 1200 CE, an intellectual flowering took place in the Middle East. Scholars translated Greek texts into Arabic, built archives, and developed large collections of manuscripts and art. Islamic collectors often donated objects for the public good, creating a cultural precedent for museums as shared repositories of knowledge and beauty.

When these Arabic texts and ideas were translated into Latin in the 12th and 13th centuries, they inspired Europe's admiration for classical antiquity. This led to a surge of private collections, funded by popes, princes, and scholars eager to preserve art and artifacts from ancient times.

During this period, the “Cabinets of Curiosities” (or Kunst- und Wunderkammern) emerged. These were private rooms filled with rare and unusual objects, both natural and man-made. They might contain sacred art, ancient fossils, “griffin’s eggs,” or relics of saints. These collections reflected a desire to understand the divine order of nature through the wonders of the world.

As one chronicler put it, the true reason behind these cabinets was a “restless desire to establish a continuity between art and nature, to demonstrate the existence of a supreme unifying principle.”

The Renaissance and the Birth of Public Museums

The word “museum” first appeared in Europe in the 15th century. A major shift occurred when wealthy families began opening their private collections to the public. The Medici family in Florence were pioneers: by the late 16th century, they opened their art and antiquities collections for public viewing. This marked the beginning of museums as places meant to share knowledge and culture, not just store treasures.

The Enlightenment and the Modern Museum

The 18th century, known as the Age of Enlightenment, transformed museums into institutions of education and civic pride. This was an era that valued reason, science, and the democratization of knowledge. One of the first truly public museums was the British Museum, founded in 1753 and opened to the public in 1759. It was established by Act of Parliament and funded by a public lottery, making its mission clear: to make knowledge and culture accessible to all.

The British Museum’s example spread rapidly. The Louvre in Paris, originally a royal palace, became a public museum in 1793 during the French Revolution. The Uffizi Gallery in Florence, designed as government offices, was opened to visitors in 1765.

These institutions didn’t just preserve art, they became centers of learning and cultural exchange, embodying Enlightenment ideals of education, access, and intellectual freedom.

The Origins of the Naturmuseum Senckenberg

In 1763, German physician and philanthropist Johann Christian Senckenberg donated his entire fortune, 95,000 guilders, to support a community hospital and scientific projects. After his death in 1772, a group of 32 Frankfurt citizens founded the Senckenberg Nature Research Society (Senckenberg Gesellschaft für Naturforschung) in 1817.

The Naturmuseum Senckenberg was founded soon after, in 1821, beginning with donations from early collectors like Johann Georg Neuburg, who contributed bird and mammal specimens. The museum moved into its landmark building on Senckenberganlage in 1907.

A fascinating moment in its history came in 1896, when the museum's mummified Egyptian child (inventory number ÄS 18) became the first mummy ever X-rayed. During World War II, the Senckenberg building was partially destroyed, but the collections had been safely evacuated beforehand, allowing the museum to survive and rebuild in the postwar years.

3. How Museums Shape People and Society

Museums don't just preserve objects, they shape how people think, feel, and connect with one another. Their impact spans education, culture, community, economics, and even mental health. In the 21st century, they've evolved from quiet halls of artifacts into active spaces for learning, well-being, and social dialogue.

Education and Lifelong Learning

Museums are some of the most powerful informal learning environments in the world. They support both children and adults in discovering new ideas through experience, not just information.
According to the Institute of Museum and Library Services (IMLS, 2023), 76% of U.S. museum visitors said their visit helped them "learn something new" or "see things differently."

In the UK, the Science Museum Group reaches more than one million students annually through its STEM programs, providing hands-on science education that complements classroom learning.
Museums make learning engaging through storytelling, interactive exhibits, and spaces that encourage curiosity. This is why many schools integrate museum visits into their curriculum, not only to teach content but to inspire creativity and critical thinking.

Cultural Identity and Heritage Preservation

Museums preserve cultural memory, they hold the tangible and intangible threads that define who we are. They safeguard art, language, artifacts, and traditions for future generations. The National Museum of African American History and Culture in Washington D.C., for instance, strengthens identity and pride among African American visitors while also educating others about history and equality.
UNESCO recognizes museums as "key institutions for safeguarding intangible heritage." They help communities maintain traditions and values even as societies modernize.

This act of preservation is deeply human: it provides a sense of belonging and continuity, reminding people where they come from and how culture connects across time.

Community Building and Social Cohesion

Modern museums often function as community hubs rather than elite spaces. They bring people together across backgrounds through shared experiences, concerts, workshops, local festivals, and open dialogues.
The Brooklyn Museum's "First Saturdays" program is a well-known example. Each month, thousands of visitors come for free art, music, and discussions, a celebration that strengthens community bonds.
According to the MOMSI Project (Measurement of Museum Social Impact, 2023), 72% of participants reported feeling "more connected to others" after a museum visit.

Museums act as safe spaces for dialogue, where people can discuss history, identity, and difficult issues. The International Council of Museums (ICOM) now defines museums as "spaces for dialogue rather than temples of knowledge."

Cognitive, Emotional, and Psychological Benefits

Museums don't just teach, they can heal and inspire. Seeing real works of art stimulates the brain far more strongly than viewing reproductions. A study at the Mauritshuis Museum in The Hague found significantly higher neurological responses when people viewed original artworks rather than copies.

Art museums have also been shown to reduce stress. A study in Museums & Social Issues revealed measurable decreases in psychological and physiological stress indicators after art-viewing sessions. The Barber Institute of Fine Arts found that museums support mental well-being and social connection by providing aesthetic experiences and emotional narratives that resonate with visitors. Museums create what psychologists call "affective engagement", moments of wonder, empathy, or reflection that linger long after the visit.

Economic and Tourism Impact

Museums are also major players in the economy. They generate jobs, attract visitors, and revitalize cities.
The Louvre Museum contributes over €1 billion annually to the Paris economy.
Research from the European Group on Museum Statistics (EGMUS, 2020) shows that every €1 invested in museums can return €3–€4 through

tourism and related services.

The phenomenon known as the “Bilbao Effect”, sparked by the opening of the Guggenheim Museum Bilbao, demonstrates how a single cultural institution can transform an entire city’s identity and economy.

Innovation and Digital Inclusion

The digital era has redefined accessibility. During the COVID-19 pandemic, museums worldwide turned to virtual exhibitions and online learning. The Rijksmuseum in the Netherlands offered high-resolution virtual tours that reached millions of users globally.

According to UNESCO (2021), 90% of museums launched digital programs during the pandemic, expanding access for rural and disabled audiences.

This digital shift didn’t just fill a gap, it permanently changed how people experience culture. It blurred the line between physical and virtual visits, ensuring that anyone, anywhere, can engage with art and heritage.

Intergenerational Connection

Museums also connect generations. They provide shared experiences for families and communities.

The Smithsonian Institution’s Family Learning Programs, for example, are designed for grandparents and children to explore exhibits together, encouraging storytelling across generations.

The American Alliance of Museums (AAM, 2022) found that intergenerational visitors report higher satisfaction and stronger family bonds.

These interactions build empathy and continuity, ensuring cultural values are passed forward.

Civic Engagement and Social Awareness

Museums are increasingly platforms for activism and dialogue. They raise awareness about climate change, equality, and justice, helping visitors see their place in a shared global story.

The Natural History Museum in London launched “Our Broken Planet”, an exhibition on climate change that inspired young visitors to take part in sustainability projects.

The Museums Association UK (2022) reported that over 60% of British museums now include social justice or environmental themes in their displays. By doing this, museums encourage visitors to think critically and act consciously, shaping societies that are better informed and more empathetic.

Museums today are no longer silent halls of artifacts, they are living, breathing participants in society, teaching, connecting, healing, and inspiring people in countless ways.

4. Museums, Art, and Design

Museums and art have always been intertwined. While artists create, museums preserve and present, shaping how art is seen, valued, and understood. The relationship between art, design, and museum spaces is dynamic, influencing everything from creative innovation to public engagement.

Museums as Platforms for Art

Museums are vital platforms for artists to present their work, experiment, and reach audiences. Many artistic movements, from Impressionism to Modernism, gained recognition largely through museum exhibitions that gave visibility and legitimacy to new forms of expression.
Without these institutions, many creative revolutions would have remained in obscurity. Museums give artists space to explore ideas and help the public encounter those ideas in meaningful ways.

The Curator's Role: Shaping Meaning Through Display
Every exhibition tells a story, and that story is shaped by curatorial decisions. What is shown, what is left out, how pieces are sequenced, and how they are contextualized all determine how visitors interpret art.
Curators act as both editors and storytellers. Their choices affect not just what people see, but what they understand about art, culture, and history.

For example, changing the order of artworks in a gallery can alter the emotional or intellectual journey of a visitor. Even something as subtle as lighting or spacing can shift perception, showing that design is part of interpretation.

Architecture and Spatial Design

The design of museum buildings themselves has become an art form. From the classical halls of the Louvre to the futuristic curves of the Guggenheim Bilbao, architecture influences how art is experienced.
Research on museum architecture and visitor engagement shows that spatial layout, lighting, and movement flow deeply affect how people emotionally connect to exhibits.

Architects and designers create environments that help visitors slow down, focus, or feel awe. A well-designed gallery guides people naturally through exhibits, using transitions between rooms to signal shifts in theme or mood.
These physical spaces are not neutral, they are part of the storytelling process. Architecture, in this way, becomes a silent collaborator with the curator and the artist.

Interactive and Emotional Design

Modern museums increasingly use interactive and digital design to create deeper, more emotional connections with audiences.
Projects like “Sensitive Pictures” at the Munch Museum explore how emotional interaction design can make art feel more personal and relatable, using technology to trigger emotional responses.

Similarly, hybrid installations, VR experiences, and participatory exhibits invite visitors to become co-creators, turning passive viewing into active engagement.
In this new model, museums aren’t just showing art; they’re designing experiences that blur the line between artist and audience.

Design as a Bridge Between Art and Understanding
Design in museums is more than decoration, it’s communication. The architecture, layout, and interactivity work together to help people make sense of art and ideas.
This is why museum design often balances aesthetic beauty with functional clarity, creating spaces that are both inspiring and accessible.
By shaping how visitors move, see, and feel, design transforms the museum from a building into a living conversation between art and people.

In every sense, museums are both keepers and interpreters of art and design. They preserve creativity from the past and provide fertile ground for the innovations of the future.

5. The Modern Museum: Technology, Inclusion, and Design

In the 21st century, museums are evolving rapidly. They are no longer just physical buildings but dynamic spaces where digital technology, accessibility, and human-centered design work together to create inclusive and engaging experiences for everyone.
From digital interactives to data-driven crowd management, today's museums combine innovation with empathy, making culture more open, equitable, and connected.

Inclusive and Accessible Digital Interactives

A modern museum experience often begins with a screen, a touchpoint, or a mobile app. But behind these tools lies a growing commitment to inclusive and accessible design.
The Smithsonian Institution has been a leader in this area through its publication "Inclusive Digital Interactives: Best Practices + Research." This resource focuses on designing digital interactives that work for visitors and staff of all abilities.

The Smithsonian's Access Initiative promotes what it calls human-centered design, a way of creating museum technology that starts with one simple question: Is this design usable for everyone?

As the Smithsonian explains: *"If we all start to apply that simple shift of perspective when asking, 'Is this design usable?' then we move to a mode of working that really works for everyone."* The Smithsonian's Guidelines for Accessible Exhibition Design add practical rules, for instance, interactives must be usable from a wheelchair-accessible height and must not block movement for people using assistive devices.

This approach ensures that every visitor, regardless of physical or cognitive ability, can explore exhibits independently and comfortably.
The idea goes beyond physical access. Inclusive digital interactives are also designed for emotional accessibility, ensuring that users connect with exhibits on a personal level. A chapter in the Smithsonian's publication describes this as "emotional accessibility in digital learning experiences and its relation to Universal Design."

Digital interactives, when designed inclusively, become tools for bridging inequality. The project "Bridging the Gap: Using Digital Interactives for Social Museums" emphasizes that digital interfaces can "eliminate inequalities and increase inclusiveness."

Data-Driven Flow Management: How Technology Guides Visitors

Technology isn't just transforming exhibits, it's changing how visitors move through museums. Managing large crowds and ensuring comfort has become an essential design challenge, especially for world-famous institutions.
At the Uffizi Gallery in Florence, researchers developed a predictive and prescriptive queue management system. The system calculates how many visitors can enter every 15 minutes, based on real-time data about how long people spend inside.
Their open-access paper explains that each day is divided into 37 slots of 15 minutes each, which also match the time slots available for online ticket booking.

To make the experience smoother, as of May 2024, the Uffizi moved to a fully digital ticketing system, turning online purchases into QR codes that visitors can scan at entry, eliminating paper queues. Their booking platform now requires visitors to select both the day and entry hour, balancing flow while reducing wait times.

Smart Museums and IoT Tracking

Other museums are taking visitor management even further with digital sensors and data modeling.
At the Galleria Borghese in Rome, researchers have created stochastic digital twins, simulations of visitor movement inside the museum. These "digital copies" help optimize crowd comfort and safety. The project uses Lagrangian IoT-based tracking, a system where portable Bluetooth Low Energy (BLE) beacons are carried by visitors and detected by fixed Raspberry Pi receivers around the museum. This data allows curators to reconstruct visitor trajectories at room scale, helping them understand which exhibits attract attention and where congestion forms.

Similar IoT and modeling techniques have also been applied in other crowded cultural spaces, like aquariums, to reduce bottlenecks and improve the visitor experience. These technologies allow museums to move toward data-driven design, blending visitor behavior analysis with architectural and curatorial planning.

- Practical Design Tactics and Guidelines
- From all these studies and case examples, several clear design tactics emerge:
- Accessibility from the start: Accessible design must be built into exhibition planning, not added later.

- Timed entry slots: The Uffizi's model of dividing the day into 15-minute slots helps manage visitor density.
- IoT tracking for flow insight: The Borghese model uses BLE beacons and sensors to map movement patterns.
- Queue reduction through digital tickets: Replacing paper tickets with QR codes speeds entry and reduces crowding.
- Inclusive co-design: Successful digital interactives involve collaboration among leadership, curators, designers, educators, and accessibility experts from the start.
- Emotional and universal design: Digital learning experiences should consider emotional as well as physical accessibility.
- Digital interactions for inclusion: Virtual and digital elements can help museums bridge inequality by reaching audiences who may never visit in person.

The Bigger Picture: Technology Serving People

In the end, the most advanced museum designs share a simple philosophy: technology should enhance human connection, not replace it. Whether through digital interactives that make learning inclusive, or sensor data that helps visitors move freely and comfortably, the best modern museums use innovation to serve people, to make art, science, and history more accessible and engaging for everyone.

6. The Senckenberg Example: Science, Society, and Sustainability

The Senckenberg Society for Nature Research and its museum in Frankfurt are excellent examples of how a modern museum can combine science, education, and social responsibility. The Senckenberg model shows that a museum isn't just a building of exhibits, it's a living organization that connects research with real-world impact.

A Museum Born from Science and Community

The Naturmuseum Senckenberg has roots that reach deep into the Enlightenment era. Its story begins in 1763, when Johann Christian Senckenberg, a physician and philanthropist from Frankfurt, donated 95,000 guilders, his entire fortune, to establish a community hospital and support scientific projects.

After his death in 1772, a group of 32 Frankfurt citizens continued his vision by founding the Senckenberg Nature Research Society (Senckenberg Gesellschaft für Naturforschung) in 1817. Their mission was to promote research and education in natural history. The Naturmuseum Senckenberg itself was established in 1821, and it quickly became one of Europe's leading centers for natural history.

It grew from donations, for instance, early contributor Johann Georg Neuburg gave his personal collection of bird and mammal specimens. By 1907, the museum had moved into its grand new building on Senckenberganlage in Frankfurt.

A remarkable moment in its scientific history came in 1896, when the museum's mummified Egyptian child (inventory number ÄS 18) became the first mummy ever X-rayed, marking a milestone in both medical imaging and museum science. Though its building was partially destroyed during World War II, the exhibits had been safely evacuated in advance, allowing the institution to recover and rebuild after the war.

Citizen Science and Public Involvement

Senckenberg stands out for how it brings the public into the process of science itself. Through citizen science programs, the museum invites everyday people to help collect and analyze data, contributing to biodiversity and environmental research. These projects blur the boundary between expert and visitor, turning curiosity into real scientific contribution.

This model strengthens public understanding of science while expanding research capacity. It also deepens trust between scientists and society, an increasingly important goal in today's world.

Educational Outreach and Youth Programs

Senckenberg is committed to education and access, especially for young people and communities that might not normally engage with science museums. Its outreach programs focus on sparking curiosity in the natural sciences through workshops, interactive lessons, and family activities. Many of these programs are tailored for children and youth with limited access to cultural opportunities.

The aim is not only to teach facts about biodiversity and evolution but to inspire lifelong learning and environmental awareness.

Sustainability and Environmental Responsibility

Sustainability is built into Senckenberg's operations and philosophy. The organization views itself as part of a larger ecological system, emphasizing the responsible use of natural resources in both its research and daily practices.

Senckenberg's mission focuses on the long-term preservation of biodiversity and natural heritage. It seeks to balance environmental protection, economic performance, and social responsibility, a triple foundation for sustainable practice. In this sense, the museum embodies what modern institutions are striving for globally: a commitment not just to studying the natural world, but to protecting it.

Inclusivity and Diversity

The Senckenberg Diversity and Inclusion Committee ensures that the institution remains open, fair, and equitable. Its initiatives promote equality, integration, and work-life balance, recognizing that a diverse workforce strengthens scientific creativity and social understanding. This inclusive culture extends beyond staff to the visitor experience, helping create an environment where everyone feels represented and valued.

Public Engagement and Dialogue

Senckenberg's mission goes beyond exhibitions. Through public talks, special events, and science festivals, it creates open spaces where researchers and citizens can exchange ideas about urgent global issues, from climate change to species extinction. These conversations help make complex research accessible to everyone, empowering communities to see how scientific knowledge relates to their daily lives.

Science and Society in Harmony

The Senckenberg model illustrates how museums can bridge the gap between academia and society. By combining public engagement, education, sustainability, and inclusivity, it proves that scientific institutions can also be agents of social and environmental progress. Senckenberg's approach echoes a broader trend in museum culture, one that sees these institutions not as repositories of the past, but as partners in building a sustainable, informed, and inclusive future.

7. The Future of Museums

Museums have come a long way, from ancient collections of curiosities to vibrant, inclusive, and data-driven centers of culture and learning. Looking ahead, their evolution continues, shaped by technology, sustainability, and a deeper commitment to serving communities. The future of museums is not only about what they display, but how they connect people, ideas, and environments.

Digital Transformation and Global Access

The digital revolution has permanently expanded what a museum can be. With the rise of virtual tours, 3D scanning, and immersive storytelling, museums now reach audiences far beyond their walls. During the COVID-19 pandemic, nearly every museum worldwide launched online initiatives to stay connected with the public. According to UNESCO's 2021 report, 90% of museums developed digital programs, vastly increasing access for rural, disabled, and international audiences.

Institutions like the Rijksmuseum in the Netherlands demonstrated how digital exhibitions can become global experiences, offering ultra-high-resolution virtual tours that allow people to explore every brushstroke of a Rembrandt from their own homes.

But this digital expansion also challenges museums to maintain the authentic emotional impact of physical experiences. As studies show, real artworks stimulate the brain more powerfully than reproductions, a reminder that even in a digital age, physical encounters with art remain irreplaceable.

The most successful institutions will likely blend both, combining in-person immersion with digital inclusion, ensuring access for all while preserving the magic of the real.

Inclusivity and Social Responsibility

Museums of the future will continue to move away from being “temples of knowledge” and toward becoming platforms for dialogue. As the International Council of Museums (ICOM) puts it, museums must be “open to the public, accessible and inclusive, fostering diversity and sustainability.” This means designing programs that reflect the voices of underrepresented communities, integrating social justice themes, and offering equal access to learning and cultural experiences. The Barber Institute of Fine Arts notes that museums already contribute significantly to well-being and social cohesion, helping people feel more connected and valued.

In the coming years, more museums are expected to adopt inclusive design standards, multilingual interpretation, and co-created exhibitions, giving communities direct input into how their stories are told.

Sustainability and Environmental Awareness

Museums are also embracing their role in the fight for environmental sustainability. Exhibitions on climate change, biodiversity, and conservation are becoming central to their missions. The Natural History Museum in London, for example, created its “Our Broken Planet” exhibition to inspire visitors to take personal and collective action for sustainability.

Meanwhile, institutions like the Senckenberg Museum are modeling sustainable practices, balancing scientific research with social and environmental responsibility.

Museums of the future may operate as green buildings, using renewable energy, recycling systems, and carbon-conscious exhibition design. They’ll likely measure not only visitor numbers but also ecological impact, ensuring that education and preservation go hand in hand with sustainability.

Technology and Human Connection

As technology becomes more sophisticated, from AI curation to IoT visitor tracking, the challenge for museums will be to use it without losing the human touch. The Uffizi Gallery’s digital ticketing and flow management systems show how data can improve visitor comfort and accessibility.

At the same time, projects like the Smithsonian’s Inclusive Digital Interactives demonstrate that innovation must serve empathy, making experiences more emotionally and physically inclusive.

The future museum, then, is both technological and personal, a place where artificial intelligence, sensors, and virtual tools enhance, rather than replace, human experiences of wonder, curiosity, and connection.

Museums as Catalysts for Change

Perhaps most importantly, museums are emerging as active participants in solving global challenges, from climate change to cultural polarization. By presenting research, art, and stories that provoke reflection, they help people imagine better futures. They can be meeting places for citizens, scientists, artists, and activists, all sharing the goal of understanding and improving the world.

Museums will increasingly act as cultural laboratories, where communities experiment with new ideas about identity, justice, and sustainability.

In Summary The museums of tomorrow will be:

- Digitally open, blending physical and virtual experiences.
- Socially inclusive, representing diverse voices and communities.
- Environmentally responsible, aligning with global sustainability goals.
- Emotionally intelligent, using design and technology to nurture empathy.
- Civically engaged, inspiring visitors to learn, reflect, and act.

8. More About the Senckenberg Museum

The Naturmuseum Senckenberg in Frankfurt is far more than a museum, it is one of Europe's foremost centers for natural science research. Operated by the Senckenberg Gesellschaft für Naturforschung (SGN), it unites public education with scientific inquiry. Behind its famous exhibits lies a network of research institutes where scientists study biodiversity, geology, and climate change. The collections are not just displays but working archives of natural history, containing millions of specimens used for global research on species, evolution, and ecosystems.

The museum's foundation in 1821 reflected a revolutionary idea for its time: that science should serve both scholarship and society. Today, that mission continues through field studies, laboratories, and collaborative projects that span continents. Researchers at Senckenberg publish hundreds of peer-reviewed papers each year, contributing to worldwide discussions on conservation and sustainability. This dual structure, museum and research institute, gives the Senckenberg a unique role as both a public attraction and a scientific powerhouse.

Visitors encounter this blend of wonder and knowledge every day. More than 400,000 people visit annually, including families, school groups, and travelers drawn by its iconic dinosaur skeletons, fossils, and exhibits on evolution and climate. Others come for educational programs, public lectures, and citizen science projects that invite people to contribute real data to biodiversity monitoring. Every exhibit is designed to link curiosity with inquiry, to show that research is not confined to laboratories but thrives in shared spaces of learning.

The Senckenberg also represents an international vision. Beyond Frankfurt, the society operates branches in Dresden, Görlitz, and Wilhelmshaven, each specializing in different aspects of natural history. Together they form a network that connects regional research to global challenges like habitat loss and climate resilience. In this way, the Senckenberg stands not only as a museum but as a living ecosystem of science, bridging discovery, education, and public engagement to better understand and protect life on Earth.

Appendix 2

Exhibits: A Comprehensive Research on the Three Selected Exhibits

1. Penguins

Quick Facts & Taxonomy

Penguins are flightless seabirds in the family *Spheniscidae*. Modern authorities recognize 18 living species across six genera.

Habitats & Distribution

Penguins are almost entirely Southern Hemisphere birds. They are found from the Antarctic ice to temperate islands and even near the equator (for example, the Galápagos penguin). They live in areas where marine food is productive and use land or ice mainly for breeding, resting, and molting. Their habitats include:

- Sea-ice coasts
- Rocky shorelines
- Sandy beaches
- Subantarctic islands

What Penguins Eat

- Fish
- Krill (small crustaceans)
- Squid

Foraging & Diving

Smaller penguin species often feed closer to the surface, while larger species such as king and emperor penguins dive deeper and for longer durations. Emperor penguins commonly dive around 100–200 meters. The deepest recorded dives exceed 500–560 meters. Their average dives last a few minutes, but deep dives can last much longer. These are the deepest dives recorded for any bird.

Social Behaviour & Life Cycle (Colony Life)

Many penguins are colonial breeders. They nest in dense breeding groups where calling, visual displays, and coordinated movement are frequent. Breeding strategies vary between species. For example:

- Emperor males incubate eggs through winter
- Other species share incubation duties
- Chicks may form “creches” (groups) while parents forage.

Behavioural Cues

- Vocal calls and individual recognition: penguins can identify mates and chicks by voice
- Nesting and pecking behaviours
- Preening and mutual preening between partners
- Huddling in cold-adapted species (tight, shifting groups for warmth with complex rotation dynamics)

Predators & Threats (For Realism & Tension)

Marine predators include:

- Leopard seals
- Orcas (killer whales)
- Some sharks

Land and shore predators include:

- Avian predators such as skuas
- Giant petrels (especially for eggs and chicks)

Human Interactions & Conservation

Many penguin species face major threats from:

- Climate change (sea-ice loss and changing prey distribution)
- Overfishing (reducing krill and fish availability)
- Oil spills
- Habitat disturbance
- Introduced predators on breeding islands

Human Interactions & Conservation

Many penguin species face major threats from:

- Climate change (sea-ice loss and changing prey distribution)
- Overfishing (reducing krill and fish availability)
- Oil spills
- Habitat disturbance
- Introduced predators on breeding islands

Several species are listed as threatened. Emperor and king penguins are especially climate-vulnerable due to sea-ice dependence. Other populations, such as African and Galápagos penguins, are threatened by warming, overfishing, and direct human impacts.

Real-World Examples

- Emperor colonies depend on stable fast-ice for months; ice break-up can cause mass chick mortality.
- King penguin colony declines have been reported in some locations and may be climate-linked.

Interesting / Useful Facts

- Penguins are countershaded (dark back, white belly)
- They “fly” underwater using wing-like flippers

2. Tardigrades (Water Bears)

Overview

Scientific name: Tardigrada

Common names: Water bear, moss piglet

Size: Usually about 0.3–0.5 mm (typically only visible under a microscope)

First described: 1773 by J.A.E. Goeze, named "little water bear" because of their slow, bear-like movement

Shape & movement: 8 stubby legs with claws; slow "walking" gait through thin water films

Fun fact: They've existed for over 600 million years

Habitat

Tardigrades are found almost everywhere on Earth (Deep sea to the Himalayas to Antarctica). They are most commonly found in:

- Mosses
- Lichens
- Soil
- Freshwater habitats
- Marine sediments

They can survive at extreme ranges, including:

- Altitudes above 6,000 m
- Depths below 4,000 m

Types & Diversity

There are around **1,300+ known species** of tardigrades.

They are classified based on body features such as:

- Number and structure of claws
- Skin patterns and structures
- Mouthparts and feeding anatomy

Life & Reproduction

Tardigrades can reproduce in multiple ways:

- Sexually
- Asexually (parthenogenesis)

Egg-laying behaviour:

Some species lay their eggs inside the shed cuticle, which provides protection.

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- Skin patterns and structures
- Mouthparts and feeding anatomy

Example species: *Acutuncus antarcticus*, a tardigrade adapted to Antarctic conditions.

Diet

Tardigrades feed using needle-like stylets that pierce cells and allow them to suck out the contents. Their diet depends on species:

- **Plant/algae feeders:** algae, moss cells, lichens
- **Carnivorous species:** rotifers, nematodes, other tardigrades
- **Decomposer feeders:** decaying organic matter

Predators & Threats

Despite their reputation, tardigrades do have natural predators:

- Nematodes
- Mites
- Snails
- Insect larvae
- Other tardigrades
- Fungal parasites
- Protozoans

Survival Superpowers ("Invincibility")

The Tun State (Cryptobiosis)

When in extreme conditions, tardigrades enter a survival mode.
In this state they:

- Lose almost all body water
- Curl up into a ball and stop moving

In the tun state they can survive:

- Extreme temperatures (from near absolute zero to above boiling)
- Very high pressure (greater than in ocean trenches)
- The vacuum of space (shown in space exposure experiments)
- Decades without food or water

Radiation Resistance

Some tardigrade species can survive radiation doses thousands of times higher than humans can tolerate. **How they survive is below.**

- **Dsup Protein (Damage Suppressor):** A unique protein that binds to DNA and reduces radiation-induced damage.
- **Efficient DNA repair:** After exposure, tardigrades activate strong DNA-repair mechanisms and rebuild damaged DNA faster than most organisms.
- **Antioxidant defenses:** Extra protective genes help neutralize harmful oxygen radicals.
- **Metabolic pause:** Reduced metabolism limits additional cellular damage during stress.

Summary – Why They're Unique

Feature	What It Means	Human Comparison
Tun state (cryptobiosis)	Can "pause" life completely	Humans cannot survive suspended metabolism
Radiation resistance	Survive >1000× human lethal dose	Humans die at ~5 Gy
Temperature range	Survive from −272 °C to >150 °C	Humans die near 50 °C
Vacuum & pressure	Can survive space vacuum and deep ocean pressure	Humans need protection for both
DNA repair	Repairs severe genetic damage	Humans cannot repair to this degree
Longevity in dormancy	Can survive decades without water/food	Humans survive only days

Important note: Tardigrades still suffer DNA damage. They survive because they repair it extremely well. Not all species are equally resistant, and extreme or prolonged exposure can still kill them.

Other Interesting Facts

- Tardigrades have survived in space and were revived afterward.
- They are found in some of the coldest and driest regions on Earth.
- They can survive for over 30 years in suspended animation.
- Some moss samples contain hundreds of tardigrades per gram, showing extremely high population density in the right conditions.

3. Armadillos and Pangolins

Armadillos

Armadillos are a unique group of mammals confined to the Americas. There are about **21 recognised species** in the family **Dasypodidae and/or Chlamyphoridae**. Most species live in Central and South America, and only one species, the **nine-banded armadillo (*Dasypus novemcinctus*)**, has extended its range into the southern United States.

Armadillos are characterised by a bony osteoderm **armour**, digging-adapted limbs, and a lifestyle often associated with ground-burrowing and insect foraging. They eat a variety of invertebrates, dig burrows, and influence ecosystem processes, for example via soil disturbance.

Despite being relatively conspicuous, armadillos remain poorly studied compared to many mammal groups. Large gaps remain in our understanding of their behaviour, physiology, population dynamics, and ecological responses to human-modified habitats.

From a conservation perspective, several species face threats from habitat destruction, road mortality, and human expansion. Researchers emphasise the need for integrated approaches

Pangolins

Pangolins are among the most threatened mammals globally. There are eight extant species (four in Africa, four in Asia) and all are subject to heavy pressure from illegal wildlife trade, primarily for their scales and meat. Their scales are made of keratin (the same material as human nails) and cover almost the entire body. The pangolin's defensive mechanism when threatened is to curl into a ball.

Ecologically, pangolins are specialised insectivores, feeding mainly on ants and termites, and often using long sticky tongues to probe insect nests. Their habitats vary from forest to savannah, and include both arboreal and burrow-dwelling species. However, many aspects of their biology remain poorly known, and basic data on distribution, abundance, and behaviour are weak.

On the conservation side, pangolins are estimated to be the most heavily trafficked wild mammals in the world, and illicit exploitation plus habitat loss push them toward extinction. Effective monitoring is complicated by their cryptic nature, low densities, nocturnal habits, and the difficulty of survey methods.

Recent research also reveals intriguing physiological adaptations. For example, pangolin scales may play a role in innate immune defence, not only as armour but also as reservoirs of antimicrobial compounds.

combining genetics, long-term field ecology, behaviour, and conservation to fill the knowledge gaps and inform protection.

Comparative and Research Key Points

Both armadillos and pangolins illustrate how relatively obscure mammalian groups can play unique ecological roles and face serious conservation threats.

While armadillos are better studied (though still with gaps), pangolins are extremely under-studied and under-protected, often described as “poorly known” in terms of basic biology. For both groups, human impacts (habitat change, hunting and trade, range alterations) are major concerns. For pangolins, illegal trade is by far the strongest threat. For armadillos, more subtle changes such as land-use, fragmentation, and climate also matter.

Research needs include long-term field studies, better inventory and monitoring methods, improved ecological and behavioural data, integration of genetics and population dynamics, and conservation action. For pangolins, methodological advances such as sign surveys, community interviews, and detection dogs are especially needed. An interesting physiological note is that the pangolin’s keratin scales may not only protect from predators but may also have evolved to support pathogen defence, making this a growing research frontier.

Differences Between Armadillos and Pangolins

Although armadillos and pangolins look similar at first glance, since both have protective armour and feed on insects, they are not closely related. Their similarities are an example of **convergent evolution**, meaning they evolved similar traits independently due to similar ecological pressures.

1. Taxonomy and Evolution

Armadillos belong to the order **Cingulata**, within the superorder **Xenarthra**, along with sloths and anteaters. Pangolins belong to the order **Pholidota** and are more closely related to carnivorans (such as cats and dogs) than to armadillos.

2. Geographic Distribution

Armadillos live only in the Americas, mostly Central and South America, with one species in the southern United States. Pangolins live only in Africa and Asia, with four species on each continent.

3. Body Armour Composition

The armour of armadillos is made of bony plates (osteoderms) covered by keratinised skin. The scales of pangolins are made entirely of keratin, the same substance as human nails and hair, with no bone inside.

4. Diet and Feeding

Both are insectivorous, feeding mainly on ants and termites. However, armadillos often also eat worms, small vertebrates, and even plants, while pangolins are strict myrmecophages, feeding almost only on ants and termites.

5. Reproduction

Armadillos show unusual traits such as polyembryony, where one egg divides into identical quadruplets (in *Dasypus novemcinctus*). Pangolins typically give birth to one offspring at a time and have long parental care periods.

6. Behaviour and Ecology

Armadillos are mostly burrowing and terrestrial animals, though some species can swim. Pangolins can be arboreal or terrestrial depending on the species. They climb trees and use their prehensile tails for balance.

7. Conservation Status

Some armadillo species face threats such as habitat loss and road mortality, but many are still relatively common. All eight pangolin species are threatened or endangered, mainly because of illegal hunting and trade for scales and meat.

Appendix 3

Complete 4W and H tables

W / H (Real Q)	Penguins (Key Findings)	Tardigrades (Key Findings)	Pangolins & Armadillos (Key Findings)
What 1 – What key scientific or natural-phenomenon message does this exhibit currently convey?	Penguin biology and conservation messaging	Extreme survival biology, tun state, medical relevance	Convergent evolution, ecology, and pangolin trafficking threats
What 2 – What are the main barriers preventing visitors from engaging deeply?	Dense layout, weak signage, few interactive moments, visitors focus on “cute visuals”	Abstract content, low visibility, weak hooks, wayfinding issues reduce discovery	Dense layout, few interactive moments, visitors focus on form not meaning, weak context framing
Who 1 – Who are the primary visitor groups engaging with this exhibit?	Families (kids 5–12) + school groups, tourists, occasional experts; adults guide children but skip dense text	Families + school groups likely, but engagement drops due to low visibility and abstract content	Families + tourists; visitors drawn by appearance, but rarely reach deeper conservation context
Who 2 – Who is affected by the issues that exhibits have?	Discomfort from cramped/low-light spaces; families need quick cues; some may react negatively to taxidermy	Visitors unsure about science hesitate to interact; families struggle without simple cues	Ethical visitors may react negatively to endangered taxidermy; families rely on quick cues and skip dense info
Who 3 – Who holds decision-making power over producing an immersive experience for this exhibit?	Decision power held by Prof. Gabler and Prof. Bedenk	Decision power held by Prof. Gabler and Prof. Bedenk	Decision power held by Prof. Gabler and Prof. Bedenk
Who 4 – Who is affected if this exhibit remains under-engaged?	Education outcomes drop; conservation goals weakened; penguins remain only a photo attraction	Medical/science relevance lost; public connection to current research declines	Conservation education weakened; emotional interest does not translate into learning or action

Who 5 – Who struggles most to engage with each exhibit?	Adults/teens expecting dramatic visuals disengage from small or information-heavy parts	Adults/teens struggle due to tiny, abstract samples and unclear interpretation	Visitors uncomfortable with taxidermy/endangered species feel guilt, sadness, or avoidance
Why 1 – Why is this particular exhibit a strong candidate for an immersive educational intervention?	High visual appeal makes it ideal for immersive learning about climate and survival	Strong "wow" factor (space survival, DNA repair) but needs clarity; immersion can improve comprehension fast	Already emotionally engaging; immersion can channel empathy into conservation knowledge and action
Why 2 – Why have visitors not engaged more deeply with this exhibit so far?	Photo culture + short label attention; overcrowding; weak scaffolding for deeper messages	Abstract science, weak metaphor, poor placement in corner reduces engagement	Visitors focus on "cute/armour look"; overcrowding; weak conservation prompts limit learning
Why 3 – Why is it important for the museum to invest in improving this exhibit?	Converts popular exhibit into conservation education and advocacy	Strengthens museum's science identity and public medical education value	Supports threatened-species education and converts emotion into awareness and behaviour
Where 1 – Where along the visitor path would a VR setup naturally fit without blocking circulation or competing with key visual anchors?	Must avoid bottlenecks; best near entry/side spaces; needs on-site mapping verification	Medical hall quieter at far end, so VR could fit more easily without crowding	Must avoid bottlenecks; best near entry/side spaces; needs on-site mapping verification
Where 2 – Where have other museums implemented short immersive experiences in similar exhibits, and what can we learn?	Strong precedent: VR/mixed reality used successfully for hidden worlds and science learning	Strong precedent: immersion makes invisible scale and biology understandable	Strong precedent: immersive storytelling can drive empathy and conservation learning
Where 3 – Where has this museum done better?	Museum performs best with dioramas and major anchor exhibits (dinosaurs, deep sea)	Museum performs best with dioramas and major anchor exhibits (dinosaurs, deep sea)	Museum performs best with dioramas and major anchor exhibits (dinosaurs, deep sea)
When 1 – When do most visitors come to the museum?	Best to run during school holidays, when children visit with parents and family groups dominate	Best to run during school holidays, when children visit with parents and family groups dominate	Best to run during school holidays, when children visit with parents and family groups dominate

When 2 – When to when is this experience going to be available?	Planned short run (2 days), but 1-week is better for iteration and measuring improvement	Planned short run (2 days), but 1-week is better for iteration and measuring improvement	Planned short run (2 days), but 1-week is better for iteration and measuring improvement
When 3 – When is the viewer meant to understand the full point behind the installation?	Depends on final concept and evaluation method	Depends on final concept and evaluation method	Depends on final concept and evaluation method
How 1 – How could the immersive experience transform the current learning potential?	Turns fast photo stops into learning moments; improves recall and family discussion	Visualises scale + medical relevance; increases comprehension and dwell time	Builds parent-child dialogue and creates clear takeaways
How 2 – How might we design it to accommodate current visitor behaviour patterns?	Short, striking, low-friction, child-friendly, few clear takeaways, supports flow	Short, striking, low-friction, uses metaphors to simplify science, supports flow	Short, striking, low-friction, leverages emotion into conservation framing, supports flow
How 3 – How has the museum already attempted to engage visitors?	Dioramas, audio stations, digital installations, playground-style elements	Dioramas, audio stations, digital installations, playground-style elements	Dioramas, audio stations, digital installations, playground-style elements
How 4 – How can the experience link directly back to physical specimens?	Mirror physical specimens in VR and end beside the real object to reinforce connection	Use magnification and direct mapping to specimens so digital links back to real artefacts	Use VR to connect armour/scales and conservation story back to physical displays
How 5 – What emotions can be triggered to make it memorable?	Joy and warmth from social behaviour, plus mild sadness about climate threats	Awe and surprise at microscopic resilience, plus pride when linked to science/medicine	Empathy and protectiveness about trafficking, plus admiration for armour design

Appendix 4

How Might We Questions

1. Youssef

<u>Pangolins & Armadillos</u>

1. How might we communicate the trafficking of armadillos?
2. How might we make people understand convergent evolution in armadillos and pangolins?
3. How might we understand the importance of armadillos in human society?
4. How might we understand the reason for the armadillo and pangolin ball form?
5. How might we differentiate between armadillos and pangolins?
6. How might we offer solutions to armadillo trafficking?
7. How might we convey their absence from Europe?
8. How might we use them against insects (especially dangerous ones)?
9. How might we learn from their armor?
10. How might we reduce their trafficking?
11. How might we rebuild their habitats?
12. How might we get armadillos and pangolins for the museum without trafficking?
13. How might we know their original species and evolutionary origins?
14. How might we understand their social life?
15. How might we benefit from them?
16. How might we communicate the trafficking of armadillos?

Waterbears (Tardigrades)

1. How might we convey that water bears are basically almost everywhere?
2. How might we convey the age of water bears?
3. How might we understand water bears in an evolutionary sense?
4. How might we replicate water bears’ super abilities for humans?
5. How might we show that water bears are sometimes carnivorous?
6. How might we destroy water bears?
7. How might we defeat water bears with another micro-life predator?
8. How might we get water bears to live in places they currently don’t?
9. How might we feel the water bear life (be in their shoes)?

Common Questions

1. How might we invoke a sense of wow from the exhibits?
2. How might we make the creatures feel funny or different?
3. How might we understand their importance for life or nature (what if they go extinct)?
4. How might we know their negative effects?
5. How might we behave if we had similar traits to these creatures?
6. How might we use these creatures in medical or other purposes?

2. Murad

Pangolins & Armadillos

1. How might we depict ball interactions?
2. How might we tell users about their conservation?
3. How might we outline the importance of their preservation?
4. How might we show their habitat?
5. How might we depict their interaction with predators?
6. How might we differentiate pangolins and armadillos in the exhibition?
7. How might we show they are the most trafficked mammals?
8. How might we show the effects of urbanization on extinction?
9. How might we unlock the potential of the exhibition?

Waterbears (Tardigrades)

1. How might we deal with their microscopic size?
2. How might we explain their relevance to medical science?
3. How might we represent their scientific value?
4. How might we show that water bears survive extreme conditions?
5. How might we make them look cute?
6. How might we deal with creating their habitat?
7. How might we make interactions with them?

Common Questions

1. How might we tell the visitor a story?
2. How might we get the exhibition to work with VR?
3. How might we make the user have the best experience?
4. How might we grab users' attention?
5. How might we make the exhibition enjoyable for a broad audience?
6. How might we make the scientific content interesting for non-experts?
7. How might we make users take part in the experience?
8. How might we share interesting facts with users?
9. How might we limit the experience to a maximum of 15 minutes?

3. Joshua

Pangolins & Armadillos

1. How might we not make a cruel VR experience for pangolins and armadillos?
2. How might we use their ball shape as an interaction point?
3. How might we visualize their different habitats?
4. How might we distract from armadillos' lack of traditional wow factors?
5. How might we distract from the taxidermy aspect?
6. How might we differentiate between pangolins and armadillos?

Waterbears (Tardigrades)

1. How might we convey the wow factor of water bears?
2. How might we visualize water bears?
3. How might we show how indestructible water bears are?
4. How might we show how old water bears are as a species?
5. How might we increase interest in microscopic life?

Common Questions

1. How might we use cuteness to increase traction?
2. How might we create an experience mindful of short attention spans?
3. How might we enhance what makes the exhibits special?
4. How might we provoke discussion about the exhibits?
5. How might we increase engagement with the exhibits?
6. How might we tell a story around the exhibits?
7. How might we make people interested before they enter the experience?
8. How might we interact with the surrounding environment?

4. Anjie

Pangolins & Armadillos

1. How might we make pangolins and armadillos visually engaging?
2. How might we communicate how humans disrupt their habitats?
3. How might we communicate that they are among the most trafficked animals?
4. How might we depict their different habitats?
5. How might we show how they roll into balls?
6. How might we differentiate species and sizes?
7. How might we show how they protect themselves from predators?
8. How might we show their interaction with humans?
9. How might we show dietary patterns?

Waterbears (Tardigrades)

1. How might we explore the microscopic scale of tardigrades?
2. How might we show that tardigrades live everywhere?
3. How might we show tardigrades survive severe conditions?
4. How might we show tardigrades entering the tun state?
5. How might we communicate that tardigrades have lived for over 600 million years?
6. How might we show different food sources?
7. How might we show other microscopic creatures they live among?

Common Questions

10. How might we emotionally connect users to the exhibits?
11. How might we compensate for low visitor engagement time?
12. How might we preserve the museum's research identity?
13. How might we design an experience children and families enjoy?
14. How might we spark discussion after the experience?
15. How might we make visitors spend more time with the exhibits?
16. How might we convince adults this is suitable for kids 12+?

Appendix 5

Six Thinking hats

Terraform Mars with Waterbears

Hats	Youssef	Murad	Joshua	Anjie
Yellow	The environment could fit the idea. It makes it more interesting/catchy.	I feel like we have a great amount of opportunities to contribute into our project with the given idea on the Mars	I can see how this is probable. Can see how we can see how it will work	Novel experience, time machines are fun to work on
Black	Too much work on the visual part, which is not good for our team's skillset	Environment is hard to implement into the water bear topic	Lack of clear idea, what is the actual concept interaction? And he says we have read the docs	Feel like people might not really realize WB are very, very powerful
Red	Really exciting, Visual appeal is great. Love the extra ordinary ENV... Def Make	The idea is my beloved, although, fear of environment is not fitting	There are plot holes but it feels like it can be an interesting experience	In to it and its so cool. It's mua
Green	We can sort of chemical mixing thing and use them for the plants regrowing	We can explore the idea that wasn't used before to transmit the WB in to almost inhabitable ENV	Add the water bear lab idea	We could add a timemachine, time jump between different stages of the terraformation
Blue	Youssef	Murad	Joshua	Anjie

Waterbear Lab

Hats	Youssef	Murad	Joshua	Anjie
Yellow	fits the museum, good for VR set up and short	The educational value of this experience is an open door for upcoming prototyping and might be really informative for our target groups	Fun, we have a good idea on how to do it	Easy to understand for a user. All of us are on the same page with this concept. All of us already know about it.
Black	It feels like the lab thing is not novel enough and the idea of a lab environment is overused.	might be repetitive, old people might not be engaged	The interaction could become repetitive	Has a potential to get boring if we don't really make it fun to interact or watch
Red	Love all of this idea. A good intuition that this is a winner	I LOVE Destroy the waterbears	I have a very good feeling about it. It seems very promising	like it, fear it might be boring unless we develop more
Green	Make the lab sometimes get destroyed and the waterbears stay alive	Good way to explore the exhibit. More interaction with adding a scientific employee in the background who we can interact with.	We could go further and make a button for more water bears which are matching each other	Additional Interaction: Player could feel the effect of the experiments on their avatar + eating each other
Blue	Youssef	Murad	Joshua	Anjie

Send Them Home

Hats	Youssef	Murad	Joshua	Anjie
Yellow	Lets users know about their habitats and varieties	With this experience we can really demonstrate the user how different the habitats of similar but entirely different species can be	Visually engaging. We can scale it down in a way it is doable	It's a good way to understand which habitat fits a/p and the concept has a big opportunity to be expanded
Black	Technical constraints might be. 2nd problem is that it feels pointless and lacks the direction/purpose	might be difficult regarding performance, lacks interaction-leads to lack of engagement	One Sided, Repetitive, Technical Constraints might constraint its only merit	Users might expect something more than this from the portals to worlds
Red	Kinda boring but there might be some hidden interests here.	Jain	it feels a little bit boring I don't know if other people would enjoy the interaction	Like it, fear we need to add more interactions or visuals
Green	If you do it wrong, you get sad reactions	Demonstrate difference between P & A habitats and Habitat struggles, and way they live in chosen condition is nice to show	You can have faulty space with predators inside and if you put them on the tree of this environment we get additional points	We can add a real physical sticker to invoke discussion later, sticker visuals could have a variety of content Have Gloved hands to hold the creatures so the clumsiness from VR glows will be compensated for
Blue	Youssef	Murad	Joshua	Anjie

Play as the Animal

Hats	Youssef	Murad	Joshua	Anjie
Yellow	Good gameified idea	It provides the opportunity to get to know the unique interaction between armoured protected animals with dangerous predators	Gives the user a good idea of the predators	Families and kids + adults will enjoy a bit of physical activities during the experience
Black	Not suitable for VR. Hardly integrated into VR.	might be stressing users, chance we won't be able to add variation of actions	Physically exhausting. it get the user scared by the jumpscares of predators	The HUD is not a nice way to show available HP, maybe use something else like a wrist band?
Red	Like the interactive idea in it, but doesn't feel inspiring	I feel bad for the eaten Armadillios & Pangolin :(	It feels like it could be a lot as an experience and seems kind of one sided on the interaction end. Also not a lot to learn	love it, engaging and fun to play- easy to understand
Green	Ways to fight back the predators	Following this concept, we can depict the interactions of both species in their natural environment when there is permanent threat from various things	Eating the right food can restore health	The avatar could become armadillo/pangolin shape
Blue	Youssef	Murad	Joshua	Anjie

Appendix 6

Persona sheets

	Interests • Casual console & mobile gaming • Tech & gadgets, reading about new devices • educational videos • Going to escape rooms, pop-up experiences, and light festivals	Goals • Hopes to find something memorable • Combine fun with learning something cool to talk about later • Try new tech in a serious place l • quality time with child	Pain and Frustrations • Long text panels and glass cases feel boring • Gets annoyed if VR graphics are low-quality or laggy • Hates standing in long lines • Dislikes shared headsets • Limited weekend time
Markus Stein **Age -** 43 **Occupation -** IT support technician **Education -** Informatik / IT training **Location -** Frankfurt am Main **Status -** Married, one child (14)	Motivation • Loves trying out new tech before everyone else • Sees VR as a playful break from a stressful work week • Wants experiences that are easy to show off: “Look what I tried at Senckenberg!”	Challenges • Can get slightly motion-sick in intense, fast-moving VR • Doesn’t want to spend half the visit on set-up or instructions • Impatient with technical issues • making best of his time • enjoy museum with child	Needs and Expectations • Clear sign: Topic and Duration • Simple, intuitive controls and a 10-second tutorial • Seated or mostly static experience to avoid dizziness • Headset cleaned between users, maybe disposable covers
	Tech and Social media • Newer Android phone • Gaming PC • Tried VR before • Uses Instagram, Reddit, YouTube • Comfortable with apps, QR codes, online booking	Content-Type Preferences • Simple diagrams or infographics instead of long articles • Likes interactive 3D previews • Short teaser clips or Behind the scenes	Brand and Influences • Tech YouTubers, Reddit threads, Google reviews • Brands like Sony, Samsung, NVIDIA • Friends who recommend “cool experiences” around Frankfurt

	Interests • Walking • Watching TV • Playing Smartphone Games • Gossiping • Parenting blogs • Simple DIY projects	Goals • Keep children busy • Hope the children learn something • Keep additional costs under control, dislikes paying extra for things	Pain and Frustrations • Fiddly child • dophamine • Does not think VR is for learning but for entertainment • Wearing a shared headset feels unhygienic • Worried her children will start getting addicted to expensive tech
Sabine Müller **Age -** 39 **Occupation -** Kindergarten Caretakers **Education -** Ausbildung **Location -** Germany, Hofheim **Status -** Married, two kids (12 & 14)	Motivation • Keep child busy • Spend “quality time” with children • Wants to feel like a “good mum”	Challenges • Easily gets dizzy • Easily gets body strained/ exhausted • Scared of feeling sick and then having to manage kids while unwell	Needs and Expectations • There’s a nearby seating/waiting place • Clamness • Option for her kids to do VR while she doesn’t have to • clear idea what the VR-experience will be like and what it is about
	Tech and Social media • Mid age IPhone and IPad • Facebook mainly but to stay cool- twitter and insta	Content-Type Preferences • Short, polished promo clips (10–30 seconds) • Likes some well made advertisements	Brand and Influences • Meta, Facebook, intagram • Apple • Guess, Zara

	Interests • Gaming • Memes • Porn • Godzilla • 67 • Collectable toys	Goals • See cool stuff • Make mother happy with spending time • Not get bored	Pain and Frustrations • Easily gets bored • Can't do freely what he wants • Can get loud when excited • Doesn't like being rushed or controlled while playing
Timo Hartmann **Age -** 14 **Occupation -** Gymnasium Student **Education -** Grundschule **Location -** Germany, Frankfurt **Status -** Single/ living with parents	Motivation • To experience something new in the tech/gaming field	Challenges • Holding his focus for a long time • Stay with mom	Needs and Expectations • Quick dopamine • Freedom to explore • Fun, not too scary
	Tech and Social media • Tiktok • Samsung • PS5 • Nintendo Switch • iPad for games • Watches YouTube Kids	Content-Type Preferences • Short clips • Gameplay videos • Tiktok brainrot • Streams	Brand and Influences • Twitch • Instagram • LEGO • Nintendo • Friends at school

	Interests • Drawing and coloring • Animals & nature books • Playing outside • Crafting (glue, glitter, DIY) • Magnificent century series • Disney movies	Goals • Stay close to parents • Feel safe during the museum visit • See exhibits up close • Avoid scary or overwhelming things	Pain and Frustrations • Doesn’t like wearing heavy goggles • Gets a bit nervous when she can’t see her parents • Sensitive to sudden noises • Doesn’t want to be forced or rushed
Aycan Mehmet Sultan **Age -** 12 **Occupation -** Gymnasium Student **Education -** Grundschule **Location -** xxxxxx **Status -** Single/ living with parents	Motivation • Wants to see interesting things • Stay close to parents	Challenges • Sensitive to loud noises and sudden movements • Doesn’t like wearing heavy or tight headsets • Gets anxious when something feels too real • Struggles with unfamiliar or dark environments	Needs and Expectations • Calm, gentle environment • Option to skip VR without feeling left out • Simple exhibits she can touch or explore slowly • A place to sit with a parent while others do VR • Clear explanations that something “is just pretend” • Needs reassurance
	Tech and Social media • Uses tablet only for drawing apps or kids’ audiobooks • Doesn’t like VR or intense games • Not on social media • Likes simple educational videos	Content-Type Preferences • Soft visuals • Animal videos • Step-by-step drawings • Calm, comforting content	Brand and Influences • Schleich (animal figures) • YouTube Kids drawing channels • School friends • Mother’s preferences • Disney

Appendix 7

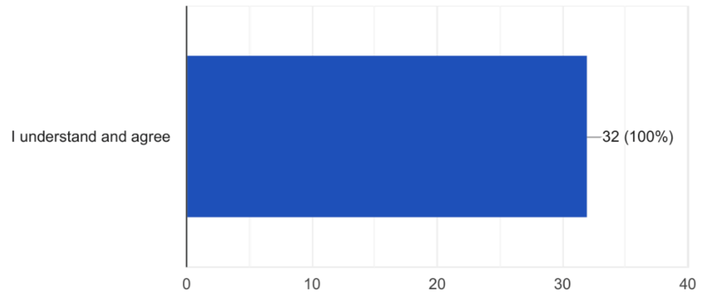
Questionnaire

Test on Microscopic life

32 responses

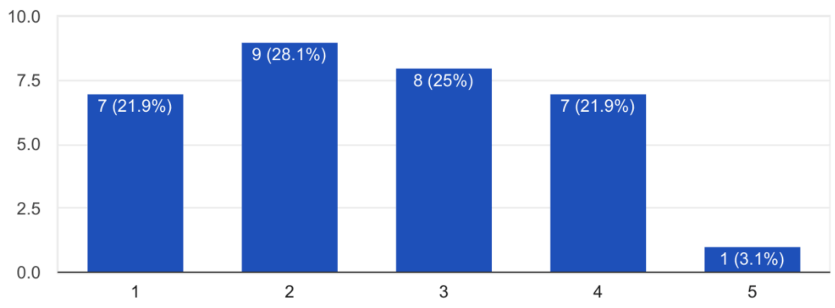
By submitting this form, I agree that the data I provide may be collected and processed by students of Hochschule Darmstadt (h_da) for educational and non-commercial project purposes.

32 responses



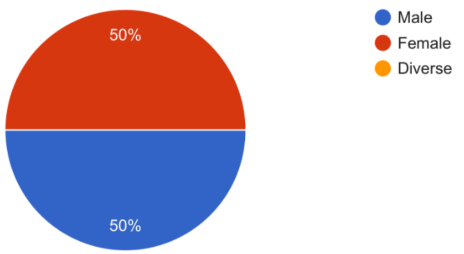
How familiar are you with VR experiences?

32 responses



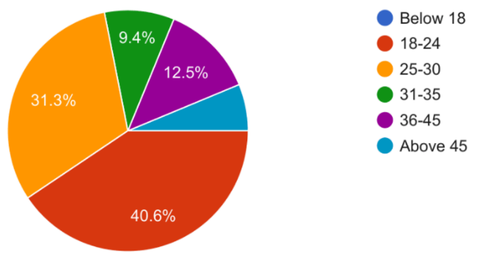
What is your gender?

32 responses



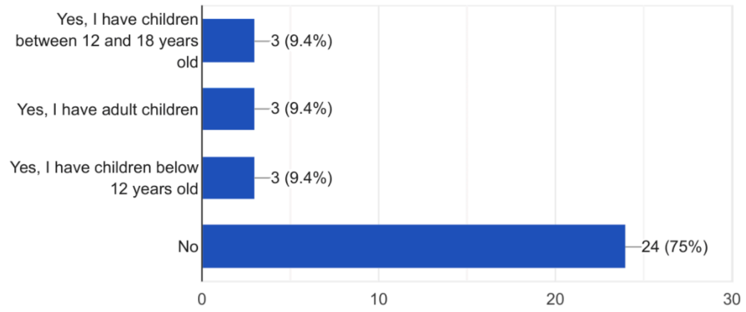
What age range are you in?

32 responses



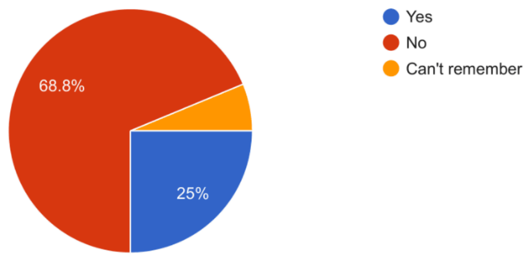
Do you have children?

32 responses



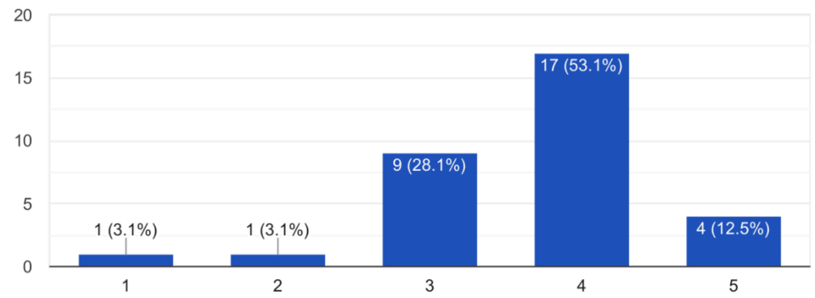
Did you already know about this VR experience before the questionnaire?

32 responses



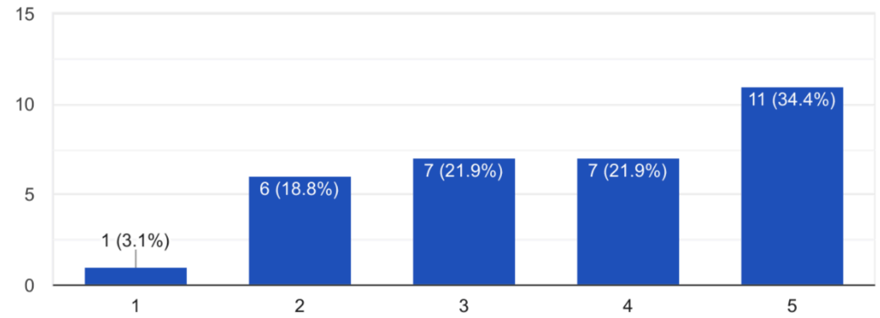
Would you enjoy an experience like this?

32 responses



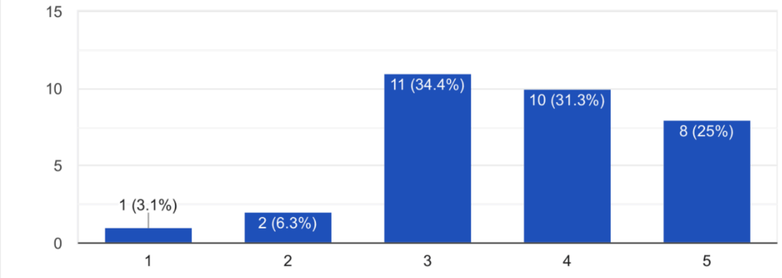
Would you enjoy the experience more if the water bear shows after each test that it is unharmed?

32 responses



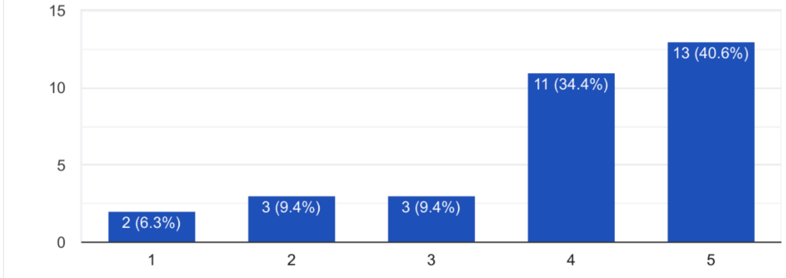
Would you feel more comfortable doing these tests if a nother character warns you when you are reaching the limits of the water bear and stops you from harming it?

32 responses



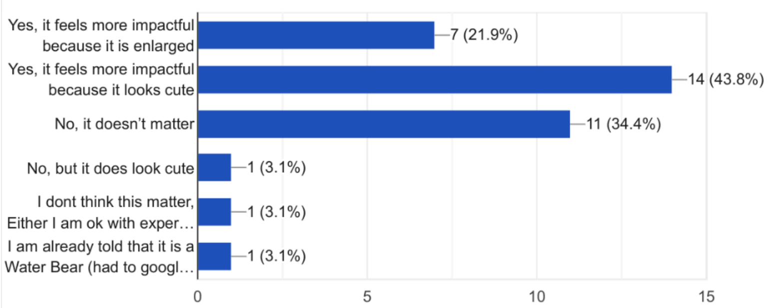
Would it make a difference to you if you were told that doing these tests helps save human lives?

32 responses



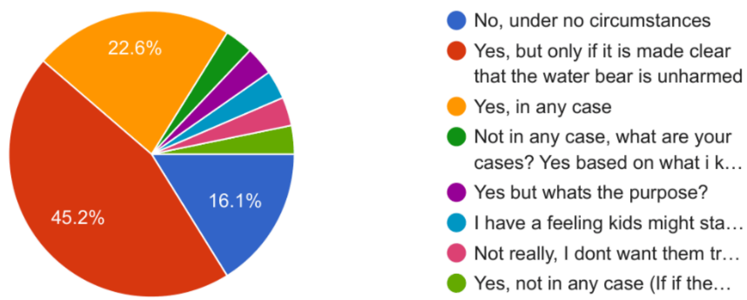
Do you think the appearance of the water bear affects your decisions?

32 responses



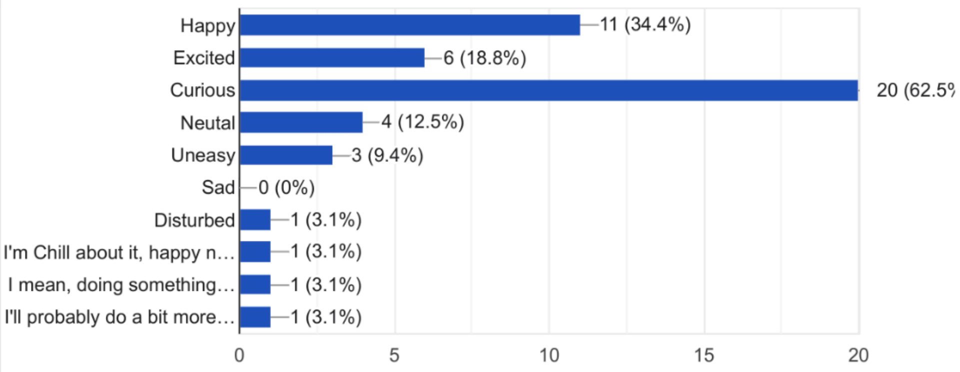
If you have (or would have) children around the age of 12–15, would you let them play this experience?

31 responses



How would you feel during the experience if it is clearly shown that the water bear isn't being harmed and that your tests help save lives?

32 responses



Appendix 8

Work Log - Complete

Start Date	Notes	Type	Anjie	Youssef	Joshua	Murad
8 October 2025	Coaching		6:00:00	6:00:00	6:00:00	6:00:00
8 October 2025	Research Museum Designs	Research	2:09:00	2:00:00	2:00:00	2:00:00
9 October 2025	Research Museum Designs	Research	0:15:00	0:15:00	0:15:00	0:15:00
9 October 2025	Process Document	Admin	1:15:00	0	0	0
14 October 2025	Consolidating all the research, and converting them to Audio for myself	Research	0:37:00	0	0	0
15 October 2025	Museum Visit	Research	8:30:00	in ppt	in ppt	in ppt
16 October 2025	Group meeting discord	Research	1:35:00	1:35:00	1:35:00	1:35:00
16 October 2025	Making the presentation structure and Adding the research in to ppt	Research	1:23:00	1:20:00	1:23:00	1:23:00
21 October 2025	Making the presentation structure and Adding the research in to ppt	Research	3:15:00	3:00:00	3:00:00	3:00:00
21 October 2025	Group meeting in Uni to polish up the Research presentation 1	Research	0:30:00	0:30:00	0:30:00	0:30:00
22 October 2025	Last review of my part and final overall formatting	Research	2:30:00	0	0	0

22 October 2025	Coaching and presenting		6:00:00	6:00:00	6:00:00	6:00:00
23 October 2025	Group meeting Discord Agree upon a some stages/stops to mark emotional journey for each exhibit Discuss with each other and mark your Journey Use the 4W and H Questions framework and fill them out for each Exhibit	Research Analysis	4:00:00	4:00:00	4:00:00	4:00:00
25 October 2025	Clean and format, proof read Emotional Journey for Presentation	Research Analysis	1:17:00	0	0	0
25 October 2025	Clean and format, proof read Emotional Journey for Presentation	Research Analysis	3:00:00	0	0	0
25 October 2025	Filling some research gaps in DT methods, and Cleaning up to a Excel	Research Analysis	2:45:00	0	0	0
25 October 2025	Quick layout for doing the 5Whys Q with group tomorrow	Research Analysis	0:11:00	0	0	0
26 October 2025	Doing a catch up on previous research we did and 5Whys in Group discord	Research Analysis	1:20:00	1:20:00	1:20:00	1:20:00
27 October 2025	Weekly report - Admin	Admin	0:24:00	0	0	0
29 October 2025	Group Working on Jam board in Class	Research Analysis	1:30:00	1:30:00	1:30:00	0
30 October 2025	Group meeting for Group constitution and slogans Discord meeting	Research Analysis	0:45:00	0:01:00	0:01:00	0:01:00
31 October 2025	Deriving sentences from Jamboard Stage 1 Discord meeting	Research Analysis	2:15:00	2:30:00	2:30:00	2:30:00
31 October 2025	Time tasking and admin	Admin	0:10:00	0	0	0
31 October 2025	Formating and cleaning up my news slide	News Flash	0:10:00	0:45:00	0:30:00	0:30:00
31 October 2025	Format and Clean up Team Constitution Doc 31 Oct Night	Research Analysis	0:15:00	0	0	0

2 November 2025	Group meeting discord final analysis on ColabBoard	Research Analysis	1:00:00	1:00:00	1:00:00	1:00:00
2 November 2025	Documentation	Admin	0	0	0	2:00:00
4 November 2025	Weekly report	Admin	0:10:00	0	0	0
5 November 2025	Doing 15 statements of HMW each individual	Ideation	0:45:00	0:45:00	0:45:00	0:45:00
6 November 2025	Branstoming Concepts as a group in cave	Ideation	2:40:00	2:40:00	2:40:00	2:40:00
8 November 2025	Adding concepts with images and text to PPT (2 Concepts)	Ideation	3:00:00	3:00:00	3:00:00	3:00:00
9 November 2025	Group meeting, going over some concepts and more brainstorming	Ideation	1:00:00	1:00:00	1:00:00	1:00:00
9 November 2025	Adding the final two remaining concepts to ppt	Ideation	0:40:00	0	0	0:40:00
9 November 2025	Admin Time sheeting	Admin	0:32:00	0	0	0
9 November 2025	Documentation	Admin	0	0	0	1:00:00
9 November 2025	Work Status report	Admin	0:20:00	0	0	0
11 November 2025	Group meet and presentation Rehearsals	Ideation	0:35:00	00:35:00	00:35:00	00:35:00
12 November 2025	Applying methods to eliminate ideas in class, eliminated 6	Ideation	4:45:00	04:45:00	04:45:00	04:45:00
12 November 2025	Discussion, eliminated 2 and Making 4 personas.	Ideation	2:25:00	02:25:00	02:25:00	02:25:00
17 November 2025	Documentation plotting and adding .1	Admin	0:10:00	0	0	0
18 November 2025	Status report and Admin	Admin	0:10:00	0	0	0

19 November 2025	Refining Concepts and choosing one concept from two using Personas One concept Chosen In class	Ideation	0:45:00	0:45:00	0:45:00	0:45:00
19 November 2025	Refining Concepts and choosing one concept from two using Personas One concept Chosen In class	Ideation	3:10:00	3:10:00	3:10:00	3:10:00
20 November 2025	Refining the chosen concept All - Thursday Morning	Ideation	2:15:00	2:15:00	2:15:00	2:15:00
22 November 2025	Admin, making folders for tasks, checking up on everyone, writing and tracking delegated tasks	Admin	0:44:00	0	0	0
22 November 2025	Making User flow	Ideation	2:01:00	2:01:00	2:01:00	2:01:00
22 November 2025	Making User flow	Ideation	1:41:00	1:41:00	1:41:00	1:41:00
23 November 2025	Reevaluation Concept again and getting on the same page	Ideation	3:35:00	3:35:00	3:35:00	3:35:00
23 November 2025	Sound Finding - Youssef	Ideation	0	2:00:00	0	0
24 November 2025	Persona testing of the Concept with Final concept, looking at completed story boards, mood board and sound	Ideation	2:10:00	2:10:00	2:10:00	2:10:00
24 November 2025	Story boarding - Joshua	Ideation	0	0	3:30:00	0
24 November 2025	CoverPage Sketch - Joshua	Ideation	0	0	0:30:00	0
24 November 2025	Documentation Murad	Ideation	0	0	0	0:30:00
24 November 2025	Presentation slide structure, initial format	Ideation	00:40:00	0	0	0
25 November 2025	Making the prototype play through acting videos	Ideation	4:30:00	4:30:00	4:30:00	4:30:00
25 November 2025	Formatting and cleaning up the Playthrough video	Ideation	0	0	4:30:00	0

25 November 2025	PPT final formatting and putting everything together	Ideation	2:14:00	0	0	0
26 November 2025	PPT final formatting and putting everything together	Ideation	8:00:00	0	0	0
26 November 2025	In class presentation and such like	Ideation	6:30:00	7:30:00	7:30:00	7:30:00
27 November 2025	Drawing WB sketch Modelling	Ideation	0	0	4:00:00	0
27 November 2025	Mindmap to Understand the Concerns we Heard in Coaching	Ideation	0	0	1:30:00	0
28 November 2025	Rigging and refining WB model	Ideation	0	0	2:00:00	0
28 November 2025	Adding to mindmap	Ideation	0	0	0:30:00	0
29 November 2025	Discussing Mindmap and concluding all the concerns	Ideation	3:15:00	3:15:00	3:15:00	3:15:00
2 December 2025	Admin	Ideation	0:10:00	0	0	0
2 December 2025	One pager text formatting and decoration	Ideation	1:20:00	0	0	0
2 December 2025	One pager image drawing	Ideation	0	0	2:00:00	0
2 December 2025	One pager Text review meeting	Ideation	1:45:00	0	1:45:00	0
2 December 2025	One pager text formatting and decoration	Ideation	0:13:00	0	0	0
2 December 2025	Questionnaire evaluation +Presentation	Ideation	0	0	4:30:00	0
2 December 2025	News Flash	Ideation	0:30:00	0:45:00	0:30:00	0:30:00
3 December 2025	Class and coaching	Ideation	6:45:00	6:45:00	6:45:00	6:45:00

4 December 2025	LabBlockout	Ideation	0	0	5:00:00	0
4 December 2025	Ideas for more interactive game	Ideation	0:30:00	0:30:00	0:45:00	0:30:00
5 December 2025	Talking about ideas discord meeting	Ideation	0	1:00:00	1:00:00	1:00:00
6 December 2025	Acting prototyping selected ideas	Ideation	3:00:00	3:00:00	3:00:00	3:00:00
8 December 2025	Character visualization Professor	Ideation	2:54:00	0	0	0
9 December 2025	Admin	Ideation	0:10:00	0	0	0
10 December 2025	Class and coaching	Ideation	7:00:00	0	0	0
12 December 2025	Brainstorm, weigh ideas, decide. Adjust the story. Dialogs – group discussion	Ideation	5:25:00	5:25:00	5:25:00	5:25:00
14 December 2025	Type story and scene plan	Ideation	2:47:00	0	0	0
14 December 2025	Type story and scene plan	Ideation	0:14:00	0	0	0
14 December 2025	Blender modelling – primitive	Ideation	0	0	03:30:00	0
14 December 2025	Coding interactions with primitive shapes	Ideation	0	10:00:00	0	10:00:00
14 December 2025	Debugging	Ideation	0	2:00:00	0	02:00:00
14 December 2025	Event / Actions system – Stage 1	Ideation	0	5:00:00	0	0
15 December 2025	Type story and scene plan	Ideation	0:06:30	0	0	0
15 December 2025	Dialogs – typing	Ideation	0:57:30	0	0	0

15 December 2025	Dialogs – typing	Ideation	1:10:00	0	0	0
15 December 2025	Dialogs – typing	Ideation	1:00:00	0	0	0
15 December 2025	Testing full interaction flow in VR	Ideation	0	0	0	00:30:00
15 December 2025	Blender modelling – unplanned	Ideation	0	0	02:30:00	0
16 December 2025	Work status report	Ideation	1:34:00	0	0	0
16 December 2026	Reading through game loop / commenting	Ideation	0	0	01:15:00	0
17 December 2025	Class and coaching	Ideation	6:30:00	06:30:00	06:30:00	06:30:00
20 December 2025	Convert Excel to Trello	Admin	1:21:00	0	0	0
20 December 2025	Test interactions, redo loops	Ideation	3:15:00	0	0	0
20 December 2025	Draw lab test equipment	Ideation	1:09:00	0	0	0
21 December 2025	Group meeting – lab equipment look, dialog jargon, character look & feel	Ideation	2:00:00	02:00:00	02:00:00	02:00:00
21 December 2025	WB-Tunstate Pose	Animation	0	0	01:30:00	0
21 December 2025	WB-Shader Placeholder	Shading / Setup	0	0	00:30:00	0
22 December 2025	Orga	Organisation	0	0	00:45:00	0
22 December 2025	WB Animation tun	Animation	0	0	00:45:00	0
23 December 2025	Admin	Admin	1:55:00	0	0	0

23 December 2025	WB Animation tun	Animation	0	0	01:00:00	0
23 December 2025	WB Animation appear	Animation	0	0	00:30:00	0
24 December 2025	WB Animation appear	Animation	0	0	01:10:00	0
31 December 2025	Waterbear appear (with issues)	Animation / Debug	0	0	01:00:00	0
2 January 2026	Waterbear appear refining	Animation	0	0	00:40:00	0
3 January 2026	Documentation	Admin	0:20:00	0	0	0
3 January 2026	Documentation	Admin	2:10:00	0	0	0
3 January 2026	Documentation	Admin	1:30:00	0	0	0
3 January 2026	Waterbear appear refining	Animation	0	0	1:00:00	0
4 January 2026	Documentation	Admin	0:20:00	0	0	0
4 January 2026	Documentation	Admin	2:00:00	0	0	0
4 January 2026	Documentation	Admin	1:10:00	0	0	0
4 January 2026	Waterbear appear refining	Animation	0	0	01:00:00	0
5 January 2026	Documentation	Admin	0:20:00	0	0	0
5 January 2026	Documentation	Admin	2:30:00	0	0	0
5 January 2026	Waterbear Idle	Animation	0	0	00:40:00	0

5 January 2026	Waterbear leave heat	Animation	0	0	00:30:00	0
5 January 2026	Test animations in Unity	Testing / Integration	0	0	00:30:00	0
5 January 2026	Music implementation and iteration	Audio	0	01:30:00	0	0
5 January 2026	3D modelling (initial pass)	Modelling	0	0	0	01:00:00
6 January 2026	Documentation	Admin	0:20:00	0	0	0
6 January 2026	Documentation	Admin	2:00:00	0	0	0
6 January 2026	Documentation	Admin	1:10:00	0	0	0
6 January 2026	Waterbear leave heat + issue	Animation / Debug	0	0	00:50:00	0
6 January 2026	Waterbear fix issue with bones tweaking	Rigging / Fixes	0	0	01:00:00	0
6 January 2026	Programming – liquid nitrogen test	Programming	0	01:30:00	0	0
6 January 2026	3D modelling (completion)	Modelling	0	0	0	01:00:00
7 January 2026	Fixing aftermath of original issue (re-keying bones)	Rigging / Fixes	0	0	01:00:00	0
7 January 2026	Waterbear leave heat	Animation	0	0	00:50:00	0
7 January 2026	Waterbear leave cold	Animation	0	0	01:00:00	0
7 January 2026	Waterbear leave pressure	Animation	0	0	00:30:00	0
7 January 2026	Music refinement	Audio	0	01:00:00	0	0

7 January 2026	Optimisation and visual improvements	Optimisation / Polish	0	0	0	01:30:00
8 January 2026	Waterbear leave radiation	Animation	0	0	01:00:00	0
8 January 2026	Redo Waterbear leave pressure (lost)	Animation	0	0	00:30:00	0
8 January 2026	Waterbear Idle Tunstate	Animation	0	0	00:20:00	0
8 January 2026	Test animations in Unity	Testing / Integration	0	0	00:40:00	0
8 January 2026	Fix animation transitions	Animation / Polish	0	0	00:20:00	0
8 January 2026	Programming – liquid nitrogen test + related logic	Programming	0	01:30:00	0	0
8 January 2026	Optimisation and making assets look good	Optimisation / Polish	0	0	0	2:00:00
9 January 2026	WB Lab desk remodeling / designing	Environment / Design	0	0	2:30:00	0
9 January 2026	Programming animations integration	Programming / Animation	0	01:00:00	0	0
9 January 2026	UV unwrapping	UV Mapping	0	0	0	01:00:00
10 January 2026	WB Lab desk remodeling / designing (+ robot arm)	Environment / Design	0	0	03:00:00	0
10 January 2026	Music polish and balancing	Audio	0	1:30:00	0	0
10 January 2026	UV unwrapping (continued)	UV Mapping	0	0	0	00:30:00
11 January 2026	WB Lab desk remodeling / designing (+ robot arm)	Environment / Design	0	0	02:30:00	0
11 January 2026	Programming – lab model changes	Programming / Environment	0	1:00:00	0	0

12 January 2026	WB Lab desk and decor	Environment / Design	0	0	02:15:00	0
12 January 2026	Programming animations tweaks and fixes	Programming / Animation	0	1:00:00	0	0
12 January 2026	Minor adjustments and fixes	Polish / Adjustments	0	0	0	00:30:00
13 January 2026	Group meeting	Production	0:45:00	0:45:00	0:45:00	0:45:00
13 January 2026	Robot Arm animation and fixing	Production	0	0	3:30:00	0
13 January 2026	Getting Blender scene unity ready	Production	0	0	0:30:00	0
13 January 2026	Setting up in unity for further use	Production	0	0	1:00:00	0
13 January 2026	Video recording of animations	Production	0	0	3:42:00	0
13 January 2026	Class	Production	3:30:00	6:00:00	6:00:00	6:00:00
14 January 2026	Edits to Voice script	Production	1:15:00	0	0	0
14 January 2026	Reading nad checking voice recording of animations	Production	0	0	0:42:00	0
14 January 2026	Editing voice script with blue	Production	0:48:00	0	0:48:00	0
15 January 2026	PressurePump Code	Production	0	3:00:00	0	0
15 January 2026	XRay Shader Effects	Production	0	4:00:00	0	0
15 January 2026	Conveyer belt + finalizing the xray machine test	Production	0	2:00:00	0	0
15 January 2026	Generation Voices	Production	4:57:00	0	0	0

16 January 2026	Petri Dish Socket lock + Kinematic movement + Win Event Trigger	Production	0	3:00:00	0	0
16 January 2026	Cold/heat/radiation shader hook to code	Production	0	2:00:00	0	0
16 January 2026	CryPodRoom outside	Production	0	0	4:15:00	0
16 January 2026	Prof Shading	Production	0	0	0:30:00	0
16 January 2026	Drawing prof	Production	0:30:00	0	0	0
16 January 2026	Drawing prof	Production	0:26:00	0	0	0
16 January 2026	Fixing Shader issues	Production	1:11:00	0	0	0
16 January 2026	Material Blending Script	Production	0:16:00	0	0	0
16 January 2026	UI screen Drawings	Production	1:08:00	0	0	0
17 January 2026	Dialog System Code + Get Back items on the table if they fall	Production	0	1:00:00	0	0
17 January 2026	Hooking up all the dialog system and mp3 files	Production	0	3:30:00	0	0
17 January 2026	Modelling and Optimising pressure pump	Production	0	0	0	2:30:00
17 January 2026	Texturing cryopods and making emission maps	Production	0	0	0	2:00:00
17 January 2026	Bake and unwrap High polly Survival room models	Production	0	0	0	1:00:00
17 January 2026	CryPodRoom outside and inside	Production	0	0	7:25:00	0
17 January 2026	UI screen Set up and testing	Production	1:42:00	0	0	0

18 January 2026	Minutes, fine tuning some variables and using the new PressurePump Model	Production	0	0:30:00	0	0
18 January 2026	Modeling + Texturing Liquid Nitrogen new Cup	Production	0	2:30:00	0	0
18 January 2026	DNA Shader	Production	0	0	0	3:30:00
18 January 2026	DNA Modelling	Production	0	0	0	0:30:00
18 January 2026	CrypodRoom+ setting up for unity, petrisish remodel	Production	0	0	8:00:00	0
18 January 2026	UI screen modelling screen, setup, adding sprites	Production	2:15:00	0	0	0
18 January 2026	Quest Issue took about three hours to trouble shoot	Production	0	0	0	0
18 January 2026	Redoing some lost work and UI screen Layout building in to unity prefab	Production	3:30:00	0	0	0
18 January 2026	UI screen Layout building in to unity prefab	Production	1:14:00	0	0	0
19 January 2026	Freezing shader, and Modelling Props	Production	0	0	0	3:30:00
19 January 2026	Modelling Props	Production	0	0	0	3:00:00
19 January 2026	Smoke effect and making alphas	Production	0	0	0	3:00:00
19 January 2026	Fixing DNA model and shader values	Production	0	0	0	1:00:00
19 January 2026	LOD Cryopod, setting up Crypod room for unity	Production	0	0	4:30:00	0
19 January 2026	Beautifying Scenes, Adding and adjusting Cryopod Scene, making SkyBox	Production	0	0	6:00:00	0
19 January 2026	UI screen Layout building in to unity prefab	Production	4:25:00	0	0	0

19 January 2026	UI screen Layout in unity building in to unity prefab	Production	2:42:00	0	0	0
19 January 2026	WB research	Production	1:05:00	0	0	0
19 January 2026	UI screen Layout	Production	2:44:00	0	0	0
19 January 2026	Scripting for Animations and Screen effects	Production	1:35:00	0	0	0
19 January 2026	Font edits UI screen	Production	0:45:00			
20 January 2026	Collecting all the timestaps from everyone and putting them together for this week	Production	0:34:00	0	0	0
20 January 2026	Screen prefab link fixed	Production	0:06:00	0	0	0
20 January 2026	Collecting all the timestaps from everyone and putting them together for this week	Production	0:58:00	0	0	0
20 January 2026	Collecting all the timestaps from everyone and putting them together for this week	Production	0:34:00	0	0	0
20 January 2026	Scene Checks, Temp Scene Change code, Petridish edits	Production	1:11:00			1:11:00
20 January 2026	Modeling Lab decor and adding material in unity scene+Particles		0	0	6:30:00	0
21 January 2026	Scene Checks, Temp Scene Change code, Petridish edits		1:11:00	0	0	0
22 January 2026	Fixing bloom issue		0	0	0:40:00	0
22 January 2026	Looking for sounds for winning		0	0	0:10:00	0
22 January 2026	Add light map, script for anim event, More visual cues for tests. UV unwarped models and replaced. Particle materials		0	0	6:30:00	0
22 January 2026	Coding the dna sample + final scene glow effect		0	3:00:00	0	0

22 January 2026	Clamp and return quick to original values when test limits reached		0	0:30:00	0	0
22 January 2026	Programming test clamps + Different player height presets switch + heat gun fixes		0	3:00:00	0	0
22 January 2026	Hookup screen to event system		0	1:30:00	0	0
22 January 2026	Documentation		1:30:00	0	0	0
23 January 2026	Bugfixing the idle audio playing after finishing		0	1:00:00	0	0
23 January 2026	Lever fix		0	1:00:00	0	0
23 January 2026	Bugfix dna done audio + tasks done event in update function + dna sample load		0	2:00:00	0	0
23 January 2026	Presentation Research and Skeleton		4:00:00	0	0	0
24 January 2026	Setting up the outline and editing the code		0	0	0	5:00:00
24 January 2026	Hours work on main menu		0	1:30:00	0	0
24 January 2026	Hour fixing heat gun		0	0:30:00	0	0
24 January 2026	Hour Haptic Feedback		0	1:30:00	0	0
25 January 2026	Fixup petridish locker position + player startup height		0	0:30:00	0	0
25 January 2026	Modelling magnifier and uv unwrapping		0	0	0	2:00:00
25 January 2026	Fixup petridish lock position again due to merge conflict		0	0:30:00	0	0
25 January 2026	Turn fog over time + Waterbear squash effect		0	1:00:00	0	0

25 January 2026	Fix clamping and fast return invoking		0	0:30:00	0	0
25 January 2026	Keep waterbear animator state on disabled		0	0:30:00	0	0
25 January 2026	Stop the petri dish a bit in the radiation machine		0	1:45:00	0	0
25 January 2026	"Find/ add sound to Cryopod scene Fix bloom/ posprocessing Change wb material to be emissive Change Color and sound nitrogen bottle and fill with liquid/smoke Find/ add celebration sound when DNA is done Add particles when stuff is inflicted to WB Adjust material of WB when inflicted WB cold animation when breath out smokes comes out of moth Robotarm sounds Precise pressure pump and fix material " Change desk model Write and add to scripts for outline Write script for cro pod scene for timed events		0	0	21:00:00	0
25 January 2026	Presentation Formatting		2:05:00	0	0	0
25 January 2026	Presentation Formatting		2:30:00	0	0	0
25 January 2026	Cryo Pod room Screen Icon drawing		1:05:00	0	0	0
25 January 2026	Cryo Pod room Screen Adding to prefab		1:38:00	0	0	0
26 January 2026	Hours attaching it to events		0	0	0	2:00:00
26 January 2026	Hook the professor win + robot win in the crypods		0	1:45:00	0	0
26 January 2026	Sounds for tools: conveyer belt, heat gun, petri dish and hologram		0	0:15:00	0	0
26 January 2026	Presentation rehearsal, final bug fixes group meetings		6:30:00	3:30:00	6:30:00	6:30:00

26 January 2026	Fixing some bugs outline shader + hooking start scene + global height adjustments		0	1:00:00	0	0
26 January 2026	Presentation Formatting		0:29:00	0	0	0
27 January 2026	quick shader fix again		0	0:30:00	0	0
27 January 2026	Presentation Formatting		5:05:00	0	0	0
27 January 2026	Admin and WorkSheet		0:52:00	0	0	0